

Qingyang Liu

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Personal Website: <https://xxxco.github.io>

Education

Duke University, School of Medicine, Durham, NC

Aug. 2024 – May 2026

Master of Biostatistics

University of Washington, Seattle, WA

Sept. 2020 – June 2024

BS in Applied Computational Mathematical Science

Experience

Data Scientist II, University of Florida School of Medicine

July. 2026 – Present

- Authored statistical analysis plan and SOP for a quality study: pilot-based power analysis, one-sided z-tests, and CIs
- Calculated sample sizes for 3 validation studies, justifying the within-record correlation assumption in methodology
- Double-programmed 832 health-plan ratings in Python and legacy SAS/R with ~2,600 automated QA checks
- Justified 4 method changes with before/after evidence, and authored methodology for state technical report

Biostatistics Research Analyst, Duke University Rohit Singh Lab

Oct. 2024 – May 2026

- Developed network-based gene prioritization framework integrating 33 biological networks across 13 categories
- Applied graph-base ranking algorithms with degree/density scoring to optimize disease-aware network selection
- Built differential expression re-ranking pipeline integrating scRNA-seq signals to improve person @10 by 100%
- Deployed Streamlit decision-support app enabling real-time rare disease gene prioritization for internal use

Statistical Analyst, Duke Center For AIDS Research

Oct. 2024 – May 2026

- Constructed maximum-likelihood phylogenetic tree via RAxML-NG under GTR+Gamma model for 584 sequences
- Evaluated transmission cluster concordance using V-measure across 7 genomic subregions and 90+ parameter sets
- Benchmarked ClusterPicker and HIV-TRACE clustering methods across genetic distance and bootstrap thresholds
- Developed R pipeline for systematic multi-method, multi-threshold sensitivity analysis of partial-genome clustering

Algorithm Engineering Intern, Ping'an Technology

July 2023 – Sept. 2023

- Processed raw video data with FFmpeg and optimized parameters to improve quality for algorithm development
- Automated analysis of 80,000+ video files in Python, extracting metadata for database management and quality checks
- Built and organized databases containing 11,000+ minutes of processed data to facilitate AI training and evaluation

Projects

Pilot Trial Designing Software, Duke University

Aug. 2024 – Mar. 2026

- Implemented a statistical framework to evaluate feasibility of pilot studies for future randomized clinical trials
- Applied Bayesian methods to estimate dropout probabilities, generate credible intervals, and assess study uncertainty
- Established composite decision rules combining acceptability and precision criteria to determine trial continuation
- Built an interactive Shiny app to calculate Bayesian assurance and support sample size planning for trial design

Multimodal Single-Cell Integration, University of Washington

Mar. 2023 – June 2023

- Built predictive models (Linear Regression, LGBM, MLP) to examine RNA-protein co-variation in single-cell dataset
- Trained models on 70,000+ CD34+ hematopoietic stem and progenitor cell samples collected from multiple donors
- Applied Python libraries (SCANPY, SHAP) to interpret outputs, quantify feature contributions, ensuring reproducibility
- Conducted single-cell RNA-seq analysis and generated visualizations to identify and communicate biological patterns

Skills

Statistical Programming: R (tidyverse, Shiny), Python, SQL, Java, SAS

Statistical & Analytical Methods: A/B Testing, Hypothesis Testing, Regression Modeling, EDA, Data Visualization

Healthcare & Bioinformatics Tools: Scanpy, SHAP, SEACells, scLink, Decipher, RaxML, HIV-Trace, TreeCluter

Research Tools: Git, Docker, LaTeX, FFmpeg